AZ41 (Mg-Al-Zn) polycrystal MD model

Build workflow up to **az41poly_with_precipitates.cfg**

INPUTS | WORKFLOW STEP: what happens inside | OUTPUTS

Online crystal database
Crystallography Open Database (CIF)

Step 1 · Download crystal structures

CIF files for the two phases were downloaded:
• hcp α-Mg (matrix) — COD **9012001**
• β-$Mg_{17}Al_{12}$ (second phase) — COD **1576218**, cubic I-43m, a ≈ **10.54 Å**

They give lattice parameters and atom positions. The Mg cell is converted to a LAMMPS **.lmp** file so Atomsk can use it as the seed for the grains.

9012001-Mghcp.cif
+ 9012001-Mghcp.lmp (hcp Mg unit cell)

1576218-Mg17Al12.cif
+ 1576218-Mg17Al12.lmp (β unit cell)

lattice data (.cif / .lmp)

Design choices
201 Å cubic box · 20 grains
target ≈ 350 000 atoms

Step 2 · Define the simulation box

```
box 201 201 201
random 20
```

param.txt.txt tells Atomsk to:
• build a **201 × 201 × 201 Å** periodic cube (≈ 350 k atoms)
• seed **20** Voronoi grain nodes at random positions with random crystal orientations

param.txt.txt

9012001-Mghcp.lmp
hcp Mg seed cell

param.txt.txt
box + grain settings

Step 3 · Build the pure-Mg polycrystal (Atomsk)

```
atomsk --polycrystal "9012001-Mghcp.lmp" param.txt.txt az41poly.lmp lmp cfg -wrap
```

• Atomsk performs a Voronoi tessellation of the box around the 20 nodes and fills each cell with hcp Mg rotated to that grain's orientation
• Atoms that overlap at grain boundaries are removed
• *-wrap* folds atoms back into the periodic box
• Every atom gets a grainID

Result: **349 469 Mg atoms** · average grain size ≈ **7.4 nm** · **No alloying yet.**

az41poly.lmp · az41poly.cfg
349 469 Mg atoms, grainID per atom

az41poly_param.txt
grain node positions + 3 rotation angles

az41poly_nodes.xsf · _grains-com.xsf
grain nodes / centres of mass

az41poly_id-size.txt · _size-dist.txt
grain size statistics

az41poly.cfg

Step 4 · Alloying to the AZ41 composition (in the .cfg)

```
atomsk az41poly.cfg -select random 13055 Mg -substitute Mg Al
-select random 1496 Mg -substitute Mg Zn az41poly_alloy2 lmp cfg
```

Random Mg atoms are swapped in place to make a random substitutional solid solution:
• Al = 349 469 × 0.037358 = **13 055** atoms
• Zn = 349 469 × 0.004280 = **1 496** atoms
• Mg = **334 918** atoms

Atom positions and grains are unchanged; only chemical identities change.

az41poly_alloy2.cfg
Mg 334 918 · Al 13 055 · Zn 1 496

az41poly_alloy2.lmp
Atomsk .lmp lacked the substitutions, so it was regenerated from the .cfg in OVITO

az41poly_alloy2.cfg

Step 5 · Insert β-$Mg_{17}Al_{12}$ precipitates on grain boundaries

```
python insert_precipitates.py
```

5a · Read inputs
• Load atoms, elements and grainID from the CFG
• Load grain rotation angles from **az41poly_param.txt**
• Expand the $Mg_{17}Al_{12}$ CIF with its **48** symmetry operators into a **58**-atom cubic cell

5b · Burgers orientation relation
$(0001)_\alpha \parallel (011)_\beta$ and $[11\text{-}20]_\alpha \parallel [\text{-}1\text{-}11]_\beta$ → rotation R_{OR}
For each host grain: $R_{lab} = R_{grain} \cdot R_{OR}$

5c · Find grain-boundary atoms
Periodic KD-tree: an atom with a neighbour within **3.6 Å** that belongs to a *different* grain is a GB atom (**~18 %** of all atoms).

5d · Pick nucleation sites
• Random GB atoms are chosen as precipitate centres
• Kept ≥ **1.6 ×** edge apart
• Continue until ~**10 vol%** is reached → **19 sites**

5e · Build & embed precipitates
• Tile the β cell, rotate by R_{lab}, crop to a **3.5 nm** cube (block shape)
• Delete host atoms inside the cube + **2.5 Å** buffer
• Remove any pair closer than **1.6 Å** (precipitate atoms win)
• Each precipitate gets grainID **101–119**

5f · Conserve composition & write
• β adds extra Al, so the nearest matrix Al atoms around each precipitate are turned back into Mg (a solute-depleted zone)
• Write the merged structure as an extended CFG

az41poly_alloy2.cfg
alloyed polycrystal (from Step 4)

az41poly_param.txt
grain orientations (from Step 3)

1576218-Mg17Al12.cif
β crystal structure (from Step 1)

Helper modules
cfg_utils.py (CFG read/write)
cif_utils.py (CIF expansion)
orientation.py (grain + Burgers rotations)

Reference paper
Fu et al., Mater. Sci. Eng. A 858 (2022) 144151
→ Burgers OR, block-shaped GB β, ~10 vol%

Tunable parameters
PRECIP_EDGE = 35 Å · TARGET_VF = 0.10
GB_CUTOFF = 3.6 Å · BUFFER = 2.5 Å
MIN_SEPARATION_FACTOR = 1.6

⚠ Scale note

The paper's β particles are ~22.8 nm, larger than the ~7.4 nm grains in this 20 nm box, so the precipitate edge is scaled down to **3.5 nm**. The Burgers OR, GB-only nucleation, block morphology and ~10 vol% are kept.

This is a **geometric construction** of the final microstructure, not a nucleation/kinetics simulation; the interfaces still **need relaxing in LAMMPS** afterwards.

merged structure

az41poly_with_precipitates.cfg
Final AZ41 model

339 897 atoms · Mg 321 521 · Al 17 100 · Zn 1 276
20 α-Mg grains + 19 Burgers-OR β-$Mg_{17}Al_{12}$ GB precipitates (~10 vol%)

Legend

Workflow step | Data file | Design choice / parameters | External source | Scripts / side note | Caveat | Final model | Thick blue arrow = step order